# E-Commerce Data Analysis

OLIST

## Overview

This report provides a comprehensive analysis of e-commerce platform performance for the years 2016 - 2018. The analysis covers revenue and sales performance , delivery & operation performance , customer behavior, seller's performance and customer retention

## Key Performance Indicator

The following metrics summarize overall business performance against the prior years

| Metric | Year 2016(1 mo) | Year 2017 | Year 2018 | YOY Change |
|---|---|---|---|---|
| Total Revenue | R$ 49.7K | R$ 6.15M | R$7.39M | ▲20.16% |
| Order Volume | 267 | 43K | 53K | ▲23.25% |
| Average Order Value | R$159.57 | R$ 138.09 | R$ 137.36 | ▼ 0.52% |

## Revenue Performance Analysis

### Years-Wise Analysis

The revenue data begin with October 2016, where the company generated a revenue of R$ 49.7K Since this represents only one month, it is not directly comparable with the subsequent years.

In 2017, **total revenue generated was R$6.15M**, showing a strong performance over the full year. In 2018, the **total revenue increased further to R$7.39M**, indicating a continued growth

### Monthly Revenue Trend

In 2017, revenue showed a consistent upward trend from January to October, **contributing 71.54% of the year's revenue**, peaking in late spring likely to be due to seasonal slowdown and possible influence of markdown inventory clearance before stabilizing in December

In 2018, revenue showed a stable trend from January to August, contributing entirely to the year's revenue, while the peak was stable in the months of March to May, likely because of the seasonal shift, followed by a plateau and subsequent decline in the rest of the year.

## Monthly revenue trend - Aggregated

After a strong start to the year that saw **50% of revenue generated by May**, the market fluctuated through a peak-and-plateau cycle before a **56% drop in September**. The recovery was historic; by November, revenue hit a record **R$1,010,271.37**, outperforming the **January start by over 740%**

Likely because of

- Conversion issues - Site downtime or checkout friction
- Inventory issues - Inability to meet or capture the demands

Yes—there are quite a few additional reasons that can cause a sharp September dip, especially in Brazil. A 56% drop usually means **multiple factors stacking together**, not just one.

➔ **Pre–Black Friday behavior**

Brazil has a **huge Black Friday culture (late November).** Most of the customers prefer to delay big purchases and often record a low conversion rate. This effect usually starts around **September/October**

➔ **Payment-cycle effects**

Most of the customers purchase the orders with installment using a credit card. Customers may still be paying off mid-year purchases, and likely to spend less.

➔ **Back-to-school**

Even though the school years start earlier. Most of the customers / families faces tuition adjustments and school supply replenishments This can create a temporary spending freeze near August.

➔ **Economics sentiment shift**

Small micro changes hit e-commerce quickly, Such as

- inflation spikes
- interest rate changes
- currency fluctuations

Brazil is quite sensitive to these → consumers become cautious fast.

### Revenue trend - Region

Across the three-year period, **São Paulo accounted for 38.2% of total revenue**, likely reflecting its significantly higher customer volume compared to other regions.

# Order Volume Trends

### Years-wise analysis

The order data begins in October 2016, during which the company completed 267 orders. As this figure represents only a single month, it is not directly comparable to the totals reported for the following full years.

In 2017, the order volume reached 43K, increasing to 53K in 2018, indicating strong year-over-year growth.

### Monthly order volume trend

The analysis reveals that monthly revenue is closely aligned with monthly order volume, suggesting a strong positive relationship between these two metrics.

# Average order value analysis

The close alignment between revenue growth and order volume indicates that AOV remained relatively stable across the period. This implies that the increase in revenue was primarily driven by higher transaction volumes rather than changes in customer spending behavior.

### Recommendations:

- **Improve conversion rates** by optimizing checkout experience and reducing potential friction points during peak months.
- **Strengthen inventory planning** to better align with demand during high-growth periods.
- **Leverage seasonal trends** by launching targeted promotions during peak revenue months (March–May).
- **Increase order volume** through marketing campaigns, as growth appears to be primarily volume-driven rather than AOV-driven.
- **Explore AOV improvement strategies** such as bundling, upselling, or discounts on higher cart values.

# Operational & Delivery Performance

In 2017, delivery performance was exceptional, with nearly 90% of orders arriving ahead of schedule. While 2018 saw a marginal 2% decline, overall efficiency remained high. However, regional disparities persist: Alagoas (AL) recorded the highest transit time at **20.25 days**—a **275.82% increase** over São Paulo (SP), which led with a 5.39-day average.

The most likely reason for longer transit time in Algos.

- Logistics networks are optimized for volume, not responsiveness. However couriers often wait to consolidate the shipments , Hence the shipment sits at hubs  more than shipment in sao paulo
- Often low-dense regions usually have fewer delivery agents and possibly larger delivery routes per agents

## Freight revenue

Total revenue includes freight charges to reflect overall business performance. However, for analytical purposes such as Average Order Value (AOV), freight is excluded to better capture actual customer spending on products.

Maranhão generated approximately R$14K in freight revenue, which stands out despite relatively low order volume. While high freight revenue can indicate strong demand across dispersed locations, in this case it likely reflects higher logistics costs per order. This suggests that customers in Maranhão are geographically spread out, leading to longer delivery distances and increased shipping expenses.

This presents a clear opportunity to improve delivery efficiency such as optimizing routes, consolidating shipments, or refining regional distribution strategies. While still maintaining service coverage and customer reach.

## Recommendation

Establishing a Lead Time and Transit Time Service Level Agreement (SLA) is essential to ensure consistency, transparency, and accountability in service delivery. By defining clear timelines for request completion, the SLA aligns stakeholder expectations and reduces ambiguity around delivery commitments.

A well-defined SLA enables better planning and prioritization, allowing teams to manage workloads efficiently while distinguishing between standard and urgent requests. It also provides a measurable framework to track performance, identify bottlenecks, and drive continuous process improvement.

Ultimately, implementing a Lead Time and Transit Time SLA minimizes delays, reduces escalations, and strengthens trust across stakeholders by ensuring predictable and reliable outcomes.

## SLA Breach rate

The SLA is established with a performance target of 90% on-time delivery, based on current process capability and historical performance

| Metric | Year 2017 | Year 2018 | YOY changes |
|---|---|---|---|
| Lead Time SLA | 9.03% | 9.96% | ▲ 0.93% |
| Transit TIme SLA | 8.32% | 10.03% | ▲ 1.71% |

## SLA Lead Time breach - By Region

In 2017, Roraima recorded the **highest breach rate at 16.67%**, representing **a 236% increase compared to Pará**. Breaches were particularly concentrated in March and November, each accounting for nearly 50% of total orders, while October saw a breach rate of 33.33%. In 2018, Roraima's breach rate declined significantly, **decreasing by 53% compared to the previous year**. Despite this improvement, April still experienced a high breach proportion of nearly 50%, while February showed a relatively lower spike at 20%.

In 2018, Pará reported a **peak breach rate of 14.56%**, which was **110% higher than that of Paraíba**. The highest concentration of breaches occurred in March, when approximately 48% of orders exceeded the SLA lead time. Although Pará's breach rate in 2017 was lower than that of Roraima, its performance lacked consistency, showing greater fluctuation compared to the more defined pattern observed in Roraima.

A key factor contributing to the high SLA lead-time breach rates in these regions is the limited seller presence. **Roraima has no active sellers**, while **Pará operates with a relatively low number of active sellers.** As the nearest operational hub, **Pará is required to support both its own demand and that of Roraima.**

This **dual responsibility places additional strain on Pará's capacity**, leading to operational inefficiencies and inconsistent SLA performance. Consequently, the increased workload and geographic dependency have contributed to elevated breach rates in both regions across the observed period. This two region contributes the increase of SLA Lead breach

## SLA Transit breach

**At 20%, March and November had the highest Transit breaches and were 844% higher than August**, which had the lowest transit breach over 2%. A potential contributing factor to this gap could be that customers are geographically spread out, leading to longer delivery distances and increased shipping expenses, to address this issue transit may intend to consolidate shipments to reduce operational cost which result in late delivery.

Inland transport delays for e-commerce shipments in Brazil during March are mainly driven by a surge in freight volumes from peak agricultural exports that strain already limited trucking capacity and road infrastructure. Heavy seasonal rains further slow down movement by damaging roads and increasing transit times, while congestion near major ports creates long queues that prevent trucks from loading or unloading efficiently. In addition, driver shortages, traffic bottlenecks around urban hubs, and occasional regulatory or customs related slowdowns contribute to reduced network fluidity, causing delays that ripple through first mile, linehaul, and last mile delivery operations.

In November, inland transport delays in Brazil are often linked to the transition into the rainy season combined with ongoing structural logistics constraints. Increasing rainfall, especially in the south and southeast, can disrupt road conditions and slow trucking flows, while storms and flooding around major corridors like São Paulo can create operational interruptions.

#### 2018 Transit SLA Deterioration

Transit SLA performance also worsened in 2018, and the underlying cause appears to be regions like Pará. Because Pará supports both local demand and nearby regions with limited seller presence, the resulting logistics strain affects both processing and transit reliability.

### Does the freight ratio affect order volume?

Customer behavior indicates that the majority of the customers are comfortable with freight ratio up to 20%, suggesting this act as a considerable threshold for shipping cost relative to order value. Beyond this threshold, customer willingness to place orders tends to decline, potentially acting as a driver for reduced conversion and lower repeat purchase rates

## Customer Behavior & Insights

### Product performance

At **3,029 units**, **Bath Table** recorded the highest order volume, which is **29.72% higher** than **Housewares**, the category with the lowest order volume at **2,335 units**.

Across the **top five product categories**, order volumes ranged from **2,335 to 3,029 units**.

### Payment preference

In 2017, Credit card payments recorded the highest frequency, which is **286% higher** than **Cash payments**. While 2018 saw a marginal lift of 1%, Overall Credit card payment remained stable and shows a growth potential.

In 2017, even though Credit card payments recorded the highest frequency, the highest money spent was **R$ 4.81K** which is **33.83% lower** than **Cash payments** and average money spent of **R$163.37**.

## Installment vs Non-Installment

At an overall **AOV of R$395**, users who opted for installments exceeding **10 cycles** recorded the highest **AOV 272.3% higher** than non-installment users. This suggests that the installment feature plays a strong role in enabling higher-value purchases and can be an effective lever for customer conversion.

However, despite their significantly higher AOV, installment users contribute substantially lower order volumes, with transactions approximately **882.39% lower** than those of non-installment users. This indicates that while installment usage drives value per transaction, it does not yet scale in terms of transaction frequency.

In terms of payment behavior, installment users show a strong preference for credit cards, which aligns with the nature of installment-based purchasing. In contrast, only **52.63% of non-installment users** choose credit cards, highlighting a more diversified payment preference among this segment.

## Basket analysis

Single-item orders show a stronger customer preference compared to multi-item orders. Additionally, in terms of Average Order Value (AOV), single-item orders outperform multi-item orders by **6.15 %.** In terms of Order Volume, Nearly 90 % of the orders are single-item orders and remaining is covered by multi-item orders

– Why do customers prefer single-item orders over multi-item orders?

- Checkout friction - A longer or complicated checkout process
- Price sensitivity and budget control - Adding more items increases the total cart value (Potential contributing factor for cart abandonment )
- Weak promotional incentives - less motivation to increase basket size

# Strategic Hypothesis: Integrated Wallet Ecosystem

### The Observation

Currently, **Installment Users** exhibit a **272.3% higher AOV** than non-installment users, yet their transaction frequency is approximately **882.39% lower**. This suggests that high-value purchases are treated as isolated events rather than part of a recurring shopping habit

### The Hypothesis

> If we implement an integrated marketplace wallet system (featuring stored balances, cashback, and partial payment options), then we will see a significant increase in

repeat purchase frequency among high-AOV users and a reduction in checkout friction for non-installment users

**Key Assumptions to Test**

- **Retention:** Stored value or rewards within a wallet will serve as a "lock-in" mechanism, increasing the likelihood of return visits.
- **Conversion:** A "Partial Payment" feature (Wallet + Card) will reduce the psychological barrier of high cart values, decreasing cart abandonment.
- **Cost Optimization:** Increased wallet usage will reduce total dependence on external credit card gateways, potentially lowering long-term transaction fees.

**Primary Success Metric (KPI)**

The success of this hypothesis will be measured by the **Growth in Repeat Purchase Rate** and a **10-15% uplift in transaction frequency** within the identified "Installment User" segment.

# Seller performance

The state of Acre demonstrates strong delivery performance, with approximately **95% of orders being delivered ahead of the estimated timeline**. However, it underperforms in terms of overall order volume, which consequently results in lower revenue generation. A potential contributing factor to this gap could be the relatively lower number of active sellers operating within the region.

While São Paulo demonstrates strong alignment between supply and demand, the rest of Brazil reveals a clear structural imbalance. High customer demand in the Northeast, particularly in Bahia, Pernambuco, Ceará, and Maranhão. Which is not matched by local seller presence, resulting in elevated freight costs, lower order volumes, and weak retention. Similarly, Northern states such as Pará and Amazonas exhibit demand under severe logistical constraints. These regions represent the most critical expansion opportunities, where increasing seller density or localized fulfillment could significantly unlock demand and improve unit economics.

**Does a region like Maranhão show lower volumes consistent with this ceiling?**

**Yes.** The data indicates that Maranhão is indeed struggling with this freight barrier because Maranhão has a very high **customer-to-seller ratio (747 to 1)** and customers are geographically spread out, the delivery distances are longer. These long distances drive up the freight cost per order, likely pushing many transactions over that **20% threshold**, which explains why the state has lower order volumes compared to its high demand.

**Marketplace Balance Analysis**

The customer-to-seller ratio indicates a [supply-constrained / oversupplied] marketplace, which may be influencing delivery timelines, pricing, and basket behavior. An increasing customer-to-seller ratio suggests growing demand pressure on sellers.

The analysis reveals a significant imbalance in the customer-to-seller ratio across regions, indicating considerable demand pressure on sellers. **Pará stands out with the highest ratio of 975**, suggesting that, on average, a single seller is responsible for serving 975 customers. **Maranhão and Piauí follow with ratios of 747 and 495 respectively**, further highlighting the strain on seller capacity in these areas.

Notably, regions such as **Amapá, Alagoas, Tocantins, and Roraima report a ratio of zero** due to the absence of sellers, despite collectively having a customer base of 807. In the absence of local sellers, demand from these regions is fulfilled by sellers in neighboring areas. This redistribution of demand further intensifies the workload on already strained sellers, exacerbating regional supply imbalances

# Customer retention analysis

## Repurchase rate analysis

The reported **0% repurchase rate** is the most critical finding in this analysis, indicating a business model currently entirely dependent on the continuous acquisition of new customers. This figure appears consistent across all segments, suggesting that the "drop-off cliff" occurs immediately following the first transaction. However, this 0% figure likely reflects a **data completeness issue**; the 2016–2018 window is relatively narrow, meaning many 2018 customers may not have had sufficient time to repeat purchase before the data collection ended. Furthermore, a lack of unique customer-level identifiers in the dataset prevents the explicit tracking of returning users, potentially masking some pockets of retention.

Due to the absence of customer-level identifiers, repurchase behavior cannot be directly measured. As an alternative, we use basket size (single vs multi-item orders) as a proxy for customer engagement.We observe that higher relative shipping cost appears to discourage basket expansion, leading to fewer multi-item purchases and lower customer engagement .

## The Customers Retention analysis

Retention is structurally broken, consistently across segments, with early drop-off, worsened by delivery failures, and only marginally influenced by data window bias.

Three key factors appear to be limiting the conversion of first-time buyers into repeat customers:

- **The "Freight Ceiling" (20% Threshold):** Customer willingness to complete a purchase declines sharply once the freight-to-order ratio exceeds **20%**. This ratio acts as a psychological barrier, directly impacting both initial conversion and the likelihood of a repeat visit.
- **Operational Instability (The March Breach):** Reliability is a key driver of retention. The **844% spike in transit breaches** during March (hitting 18.89%) represents a significant breakdown in trust. These delays, potentially caused by a widely spread customer base in specific regions, create negative post-purchase experiences that discourage future orders.
- **Basket Friction:** The preference for single items is likely a defensive behavior. Customers avoid multi-item orders due to perceived **checkout friction**, budget control concerns, and a lack of compelling promotional incentives (such as bundling rewards) to increase their basket size.

# Strategy recommendation

- Establishing region-wise Service Level Agreements (SLAs) for Lead Time and Transit Time is critical to ensure consistency, transparency, and accountability in service delivery across markets.
- Introducing a wallet feature can enhance control over the financial ecosystem while reducing checkout friction, thereby improving the overall customer experience and conversion rates.
- Launching targeted customer acquisition campaigns with compelling incentives, such as discounts and free shipping, can drive user growth. These campaigns can also be strategically designed to encourage multi-item purchases through bundled or mixed-order conditions.
- Maintaining an optimal freight-to-order value ratio is essential to sustain customer willingness to purchase, as higher ratios may negatively impact conversion and repeat purchase behavior.

# Conclusion

Overall, the analysis reveals that while the business is experiencing strong growth driven by increasing order volumes, key structural gaps remain across customer behavior, logistics efficiency, and seller distribution. Customer sensitivity to freight ratio thresholds and a strong preference for single-item orders are limiting both conversion and revenue expansion.

Addressing these challenges through optimized shipping strategies, targeted acquisition and bundling incentives, and improved checkout experience can unlock higher order volumes and repeat purchases. In parallel, strengthening SLA adherence and expanding seller presence in underserved regions will be critical to improving service reliability and supporting sustainable, scalable growth.  .